ECEN-662: Homework 7

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Problem 1

Let $X = (X_1, \ldots, X_n)^t$ where

$$X_i \overset{iid}{\sim} Bernoulli(\theta),$$

where θ is the unknown parameter. Assume that the prior distribution for Θ is $\mathcal{U}((0,1))$.

(i) Find the Bayes estimator of θ under the squared-error loss.

(ii) Now let X_{n+1} be a new observation such that $X_{n+1} \sim Bernoulli(\theta)$ and is independent of X given θ. Find $\mathbb{P}(X_{n+1} = 1|X)$.

Solution

Part (i): We know the likelihood

$$L(\theta \mid X) = \theta^{\sum_{i=1}^n X_i}(1-\theta)^{n-\sum_{i=1}^n X_i}$$

So

$$\pi(\theta \mid X) \propto \theta^{\sum_{i=1}^n X_i}(1-\theta)^{n-\sum_{i=1}^n X_i} \cdot 1$$

Which means

$$\mathbb{E}[\Theta|X] = \frac{\sum_{i=1}^n X_i + 1}{(\sum_{i=1}^n X_i + 1) + (n + 1 - \sum_{i=1}^n X_i)} = \frac{\sum_{i=1}^n X_i + 1}{n+2}$$

Which is the Bayes estimator under the squared-error loss.

Part (ii): We have

$$\mathbb{P}(X_{n+1} = 1 \mid X) = \int_0^1 \theta\pi(\theta \mid X)\, d\theta = \mathbb{E}[\mathbb{P}(X_{n+1} = 1 \mid X) \mid X] = \mathbb{E}[\Theta \mid X]$$

So

$$\mathbb{P}(X_{n+1} = 1 \mid X) = \frac{\sum_{i=1}^n X_i + 1}{n+2}$$

Problem 2

Suppose that X has a density of exponential form

$$f_\theta(x) \propto_x h(x) \exp\left(\sum_{i=1}^{d} \theta_i T_i(x) - A(\theta)\right)$$

where each $T_i(x)$ is differentiable.

(i) Show that if X has support in real line and g is any differentiable function such that $\mathbb{E}_\theta[g'(X)] < \infty$, then

$$\mathbb{E}_\theta\left[\left(\frac{h'(X)}{h(X)} + \sum_{i=1}^{d} \theta_i T_i'(X)\right) g(X)\right] = -\mathbb{E}_\theta[g'(X)].$$

(ii) Using part a), show that if $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathrm{Cov}_\theta[g(X), X] = \sigma^2 \mathbb{E}_\theta[g'(X)]$$

where $\theta = [\mu, \sigma^2]^t$.

(iii) Use part ii) to compute the third and fourth moments of $\mathcal{N}(\mu, \sigma^2)$.

Solution

Part (i): We have

$$\mathbb{E}_\theta[g'(X)] = \int_{-\infty}^{\infty} g'(x) f_\theta(x)\, dx = -\int_{-\infty}^{\infty} g(x) f_\theta'(x)\, dx$$

Where $f_\theta'(x) = f_\theta(x)\left(\frac{h'(x)}{h(x)} + \sum_{i=1}^{d} \theta_i T_i'(x)\right)$. So

$$\mathbb{E}_\theta[g'(X)] = -\int_{-\infty}^{\infty} g(x) f_\theta(x) \left(\frac{h'(x)}{h(x)} + \sum_{i=1}^{d} \theta_i T_i'(x)\right) dx = -\mathbb{E}_\theta\left[\left(\frac{h'(X)}{h(X)} + \sum_{i=1}^{d} \theta_i T_i'(X)\right) g(X)\right]$$

Part (ii): For $X \sim \mathcal{N}(\mu, \sigma^2)$, we have

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{\mu x}{\sigma^2} - \frac{x^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log\sigma^2\right)$$

Where $h(x) = \frac{1}{\sqrt{2\pi}}$, $h'(x) = 0$, $\{\theta_1, \theta_2\} = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right)$, $\{T_1(x), T_2(x)\} = \{x, x^2\} \Rightarrow \{T_1'(x), T_2'(x)\} = \{1, 2x\}$. Thus,

$$\begin{aligned}
\mathbb{E}_\theta[g'(X)] &= -\mathbb{E}_\theta\left[\left(\frac{h'(X)}{h(X)} + \sum_{i=1}^{d} \theta_i T_i'(X)\right) g(X)\right] \\
&= -\mathbb{E}_\theta\left[\left(\frac{0}{(2\pi)^{-1/2}} + \left(-\frac{\mu}{\sigma^2} - \frac{2x}{2\sigma^2}\right)\right) g(X)\right] \\
&= \mathbb{E}_\theta\left[\frac{\mu + x}{\sigma^2} g(X)\right]
\end{aligned}$$

Since $\mathrm{Cov}_\theta[g(X), X] = \mathbb{E}_\theta[g(X)X] - \mathbb{E}_\theta[g(X)]\mathbb{E}_\theta[X]$, we have

$$\sigma^2 \mathbb{E}_\theta[g'(X)] = \mathrm{Cov}_\theta[g(X), X]$$

Part (iii): Third moment:

$$\begin{aligned}\mathbb{E}[X^3] &= 2\sigma^2\mu + \mathbb{E}[X^2]\mathbb{E}[X] \\ &= 2\sigma^2\mu + (\sigma^2 + \mu^2)\mu \\ &= 3\sigma^2\mu + \mu^3\end{aligned}$$

Fourth moment:

$$\begin{aligned}\mathbb{E}[X^4] &= 3\sigma^2\mathbb{E}[X^2] + \mathbb{E}[X^3]\mathbb{E}[X] \\ &= 3\sigma^2(\sigma^2 + \mu^2) + (3\sigma^2\mu + \mu^3)\mu \\ &= 3\sigma^4 + 6\sigma^2\mu^2 + \mu^4\end{aligned}$$

Problem 3

Let

$$f_\theta(x) = h(x)\exp(\theta x - A(\theta))$$

be a one-parameter, canonical exponential family.

(i) Show that for any prior $\pi(\theta)$,

$$\mathbb{E}_\pi[\Theta \mid X = x] = \frac{d}{dx}\log\left(\frac{m_\pi(x)}{h(x)}\right),$$

where $m_\pi(x)$ is the marginal distribution of X.

(ii) Use part i) to prove Tweedie's formula:

$$\mathbb{E}_\pi[\Theta \mid X = x] = x + \frac{d}{dx}\log m_\pi(x),$$

when f_θ is $\mathcal{N}(\theta, 1)$.

Solution

Part (i): We know

$$m_\pi(x) = h(x)\int \pi(\theta)\exp(\theta x - A(\theta))\,d\theta$$

So

$$\mathbb{E}_\pi[\Theta \mid X = x] = \frac{\int \theta f_\theta(x)\pi(\theta)\,d\theta}{m_\pi(x)} = \frac{h(x)}{m_\pi(x)}\int \theta\pi(\theta)\exp(\theta x - A(\theta))\,d\theta$$

And since $\nabla p = p + \nabla \log p$, we have

$$\mathbb{E}_\pi[\Theta \mid X = x] = \frac{h(x)}{m_\pi(x)}\frac{d}{dx}\log\left(\frac{m_\pi(x)}{h(x)}\right) = \frac{d}{dx}\log\left(\frac{m_\pi(x)}{h(x)}\right)$$

Part (ii): We have

$$f_\theta(x) = \frac{1}{2\pi}\exp(-\frac{(x-\theta)^2}{2})$$

So $h(x) = \frac{1}{2\pi}\exp(-\frac{x^2}{2})$. Thus,

$$\mathbb{E}_\pi[\Theta \mid X = x] = \frac{d}{dx}\log m_\pi(x) + \frac{d}{dx}\log\left(\frac{1}{2\pi}\exp(-\frac{x^2}{2})\right) = x + \frac{d}{dx}\log m_\pi(x)$$

Problem 4

Assume that

$$\Theta_i \overset{iid}{\sim} \pi(\theta_i) \quad \text{and} \quad X_i \mid \Theta_i = \theta_i \overset{ind}{\sim} Poisson(\theta_i),$$

Show that for any prior π and any $i \in [n]$, we have

$$\mathbb{E}_\pi \left[\Theta_i \mid X_i = x_i\right] = \frac{(x_i+1)\, m_\pi\left(x_i+1\right)}{m_\pi\left(x_i\right)} = \frac{(x_i+1)\, \mathbb{E}_\pi\left[Y_{x_i+1}\right]}{\mathbb{E}_\pi\left[Y_{x_i}\right]},$$

where $Y_x = \sum_{i=1}^n 1_{\{X_i = x\}}$.

Solution

We know

$$\mathbb{E}_\pi \left[\Theta_i \mid X_i = x_i\right] = \frac{\int \theta f_\theta(x_i)\pi(\theta)\, d\theta}{m_\pi(x_i)}$$

and since $f_\theta(x) = \frac{\theta^x e^{-\theta}}{x!}$, we have

$$\mathbb{E}_\pi[\Theta_i \mid X_i = x_i] = \frac{1}{x_i! m_\pi(x_i)} \int \theta^{x_i+1} e^{-\theta} \pi(\theta)\, d\theta = \frac{(x_i+1)m_\pi(x_i+1)}{m_\pi(x_i)}$$

Since $Y_x = \sum_{i=1}^n 1_{\{X_i=x\}}$, we have $\mathbb{E}_\pi[Y_x] = n\mathbb{P}(X_i = x) = nm_\pi(x)$. Thus,

$$\mathbb{E}_\pi[\Theta_i | X_i = x_i] = \frac{nm_\pi(x_i+1)}{nm_\pi(x_i)} = (x_i+1)\frac{\mathbb{E}_\pi[Y_{x_i+1}]}{\mathbb{E}_\pi[Y_{x_i}]} = \frac{(x_i+1)m_\pi(x_i+1)}{m_\pi(x_i)}$$